Sequential Ckts:- o/p at any time depends on present i/p & past o/p. One bit memory.

Characteristic Table of flip flops.

S	R	Q_{n+1}	J	K	Q_{n+1}	T	Q_{n+1}	D	Q_{n+1}
0	0	Q_n	0	0	Q_n	0	Q_n	0	0
0	1	0	0	1	0	1	Q_n'	1	D
1	0	1	1	0	1				
1	1	X	1	1	Q_n'				

Excitation Table

Q_n	Q_{n+1}	S	R	J	K	T	D	
0	0	0	X	0	X	0	0	$T = Q_n \oplus Q_{n+1}$
0	1	1	0	1	X	1	1	$D = Q_n$
1	0	0	1	X	1	1	0	
1	1	X	0	X	0	0	1	

Designing of any f/f with known f/f.

- Design a f/f which has chara. table as given with T f/f.

A	B	Q_{n+1}
0	0	0
0	1	1
1	0	Q_n'
1	1	Q_n

Sol

A	B	Q_n	Q_{n+1}	$T = Q_n \oplus Q_{n+1}$
0	0	0	0	0
0	0	1	0	1
0	1	0	1	1
0	1	1	1	0
1	0	0	1	1
1	0	1	0	1
1	1	0	0	0
1	1	1	1	0

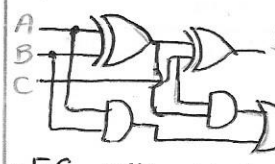
	$B'Q_n'$	$B'Q_n$	BQ_n	BQ_n'
A'	0	1	0	1
A	1	1	0	0

$T = B'Q_n + AB' + A'BQ_n'$

Combinational Ckts:-

H.A.	H.S.	F.A.	F.S.
$S = A \oplus B$	$Sub = A \oplus B$	$S = A \oplus B \oplus C$	$Sub = A \oplus B \oplus C$
$C = AB$	$B = A'B$	$C = A'B + BC + CA$	$B = A'B + A'C + BC$

- FA with H.A.

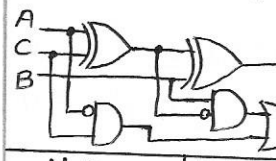


$S = A \oplus B \oplus C \; [2 * t_{pd(ex-OR)}]$

$C = C(A \oplus B) + AB$

$[t_{pd(ex-OR)} + t_{pd(AND)} + t_{pd(OR)}]$

- FS with H.S.



$Sub = A \oplus B \oplus C$

$B = A'C + [A \odot C]B$

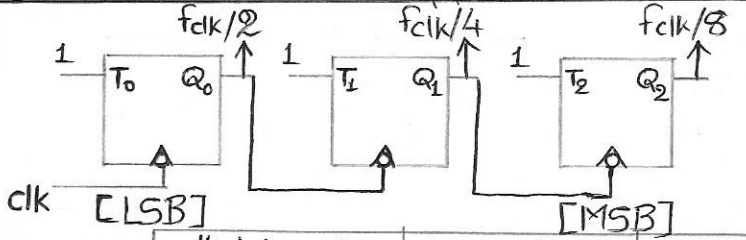
	H.A.	H.S.	F.A.	F.S.
NAND:-	5	5	9	9
NOR:-	5	5	9	9

- n-bit parallel adder/Ripple adder.

Carry propogation delay = n (one stage carry delay)

Operation delay/Total sum propogation delay = (n-1)(one stage carry delay) + 1 (sum delay of MSB stage)

Asynchronous Counter/Ripple Counter.



clk taken from	triggering of f/f	Counter
Q	+Ve	Down
Q	-Ve	Up
Q'	+Ve	Up
Q'	-Ve	Down

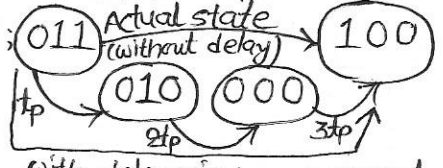
- if a single f/f provides delay of t_{pdff}.

worst delay in Asychronous counter = $n t_{pdff}$.

$\therefore t_{clk} \geq n(t_{pdff})$; n = no. of f/fs.

$\Rightarrow f_{clk(max)} = \dfrac{1}{n(t_{pdff})}$.

- Either Up (or) Down counter is possible.
- for an up counter;



with delay ripples are present.

- STROBE is used to avoid ↑

NUMBER System:-

- Base r system has numbers from 0 to (r-1).
- $(xyz.abc)_r = (x \cdot r^2 + y \cdot r + z \cdot r^0 + a \cdot r^{-1} + b \cdot r^{-2} + c \cdot r^{-3})_{10}$

Binary unsigned No. = 1010

1's compliment = 0101 [Compliment each bit]

2's compliment = 0110 [keep writing no from LSB untill first 1; then compliment]

	Unsigned	Signed	1's Compliment	2's Compliment
+0	000	000	000	000
+1	01	001	001	001
+2	10	010	010	010
+3	11	011	011	011
-0	NOT POSSIBLE	100	111	
-1		101	110	111
-2		110	101	110
-3		111	100	101
-4				100

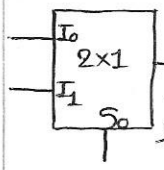
only -ve nos are complement

Range (n-bit)		$-[2^{(n-1)}-1]$ to $2^{(n-1)}-1$	$-[2^{(n-1)}-1]$ to $2^{(n-1)}-1$	$-2^{(n-1)}$ to $2^{(n-1)}-1$

Megnitude Comparator

1-bit		2-bit [$A = a_1a_0$; $B = b_1b_0$]
A>B	AB'	$a_1b_1' + (a_1 \odot b_1)a_0b_0'$
A=B	$A \odot B$	$(a_1 \odot b_1) \cdot (a_0 \odot b_0)$
A<B	$A'B$	$a_1'b_1 + (a_1 \odot b_1)a_0'b_0$

Mux:- many to one converter, parallal to serial converter, universal ckt.

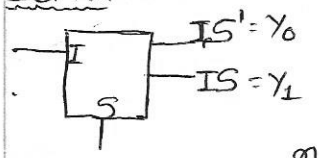


$Y = S_0'I_0 + S_0I_1$

NOT; AND; OR:- (1) 2x1 Mux.

NAND, NOR, Ex-OR; Ex-NOR:- (2) 2x1 Mux

DeMux:- Serial to parallal converter.

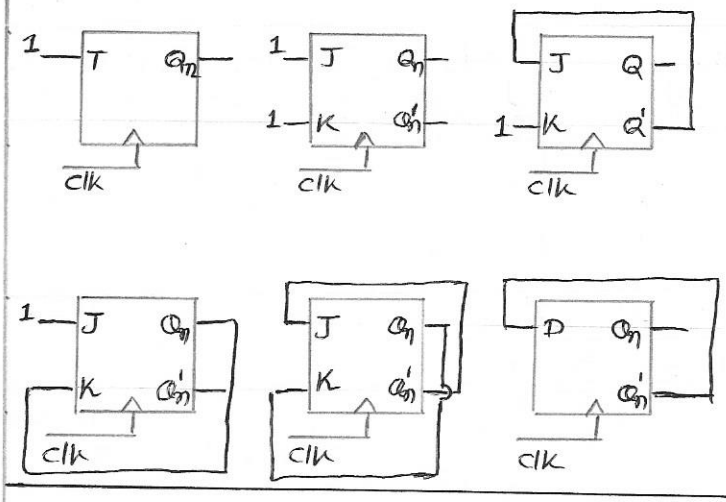


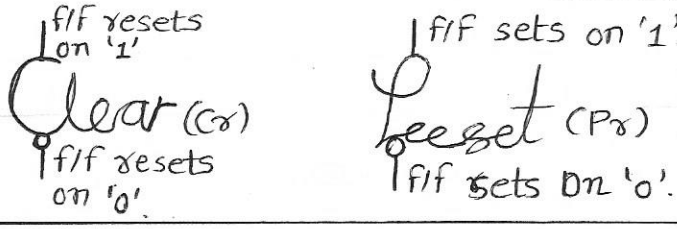
Decoder:- $n \times 2^n$
- Binary to any other code
- Used in memory
- Can be implemented with DeMux.

Encoder:- $2^n \times n$ - Any code to Binary

6

Toggle ckts/freq^n divide by 2 ckts:-





Counter:- Stable o/p states represent Mod. of a counter.

- o/p freq^n = $\frac{\text{i/p clk freq}^n}{\text{Mod}}$
- Mod-M & Mod-N in cascading = Mod(MN)

2

- **Subtraction with adder by 1's & 2's complt**

- Performing Sub in decimal s/m with 9's complt.

(5 - 4) = 5 + (9's complt of 4)
= 5 + (9 - 4) = 5 + 5 = 10 ← Carry has generated
+ 1 ← Add carry to get ans.
(1)

(5 - 4) = 5 + (10's complt of 4)
= 5 + (10 - 4) = 5 + 6 = 11 ← Carry has generated; discard it to get ans.
= (1) ☺

(4 - 5) = 4 + (9's complt of 5)
= 4 + (9 - 5) = 4 + 4 = 8 ← Carry hasn't generated
→ answer is in 9's complt & negative ∴ (-1)

Same for 10's complt

1's complt	2's complt
carry ✓ :- add carry	discard carry
carry X :- ans. is in 1's complt & '-'ve	ans. is in 2's complt & '-'ve.

- **Overflow:-** while addition ans. is out of range for specified number of bits.

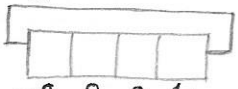
(i) $x'y'z + xyz'$ (ii) $C_{in} \oplus C_{out}$.

8

Synchronous Counters:-

- clock is common for all the f/fs.
- if for a single f/f delay is t_{pdff}; worst delay is t_{pdff}.
- Any o/p sequence can be generated.

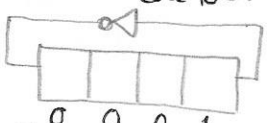
Ring Counter:- it's shift right resistor.



0	0	0	1
1	0	0	0
0	1	0	0
0	0	1	0
0	0	0	1

for n-bit ring counter → Mod-n

Johnson Counter:-



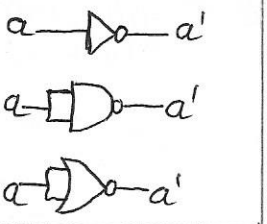
for n-bit johnson counter → Mod-(2n)

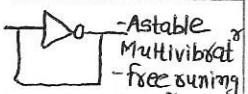
0	0	0	1
0	0	0	0
1	0	0	0
1	1	0	0
1	1	1	0
1	1	1	1
0	1	1	1
0	0	1	1
0	0	0	1

4

Logic Gates:-

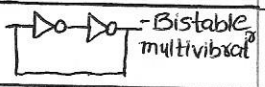
NOT [a → a']





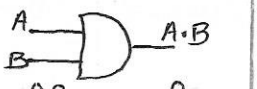
- Astable Multivibrator
- free running osc^r.

- with delay of t_p.
  o/p freq^n = $\frac{1}{2t_p}$
- if odd n number of not gates with t_p delay of each
  $\Rightarrow f_{o/p} = \frac{1}{2nt_p}$.

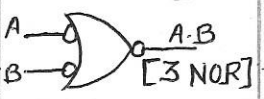


- Bistable multivibrator

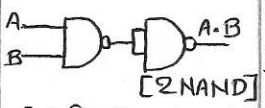
AND [Y = A·B]



00	0
01	0
10	0
11	1



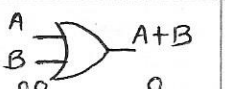
[3 NOR]



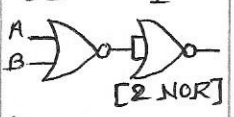
[2 NAND]

- for floating terminal prefer TTL logic family & that terminal as '1'.

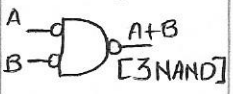
OR [Y = A + B]



00	0
01	1
10	1
11	1



[2 NOR]

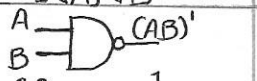


[3 NAND]

- for floating i/p prefer ECL logic family & i/p as '0'.

Universal Gates.

NAND

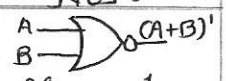


00	1
01	1
10	1
11	0

- if any one i/p is '0' ⇒ o/p = '1'

[4 NOR]

NOR



00	1
01	0
10	0
11	0

- if any one i/p is '1' ⇒ o/p = '0'

[4 NAND]

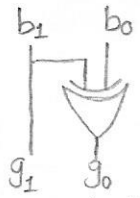
Associative prop

NAND, NOR	✗
All other Gates.	✓

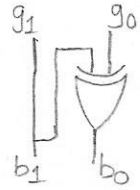
Codes:-

	BCD	Exess-3	Gray
0	0000	0011	0000
1	0001	0100	0001
2	0010	0101	0011
3	0011	0110	0010
4	0100	0111	0110
5	0101	1000	0111
6	0110	1001	0101
7	0111	1010	0100
8	1000	1011	1100
9	1001	1100	1101

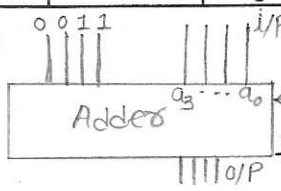
- self complementary code
- Unit distance Code.



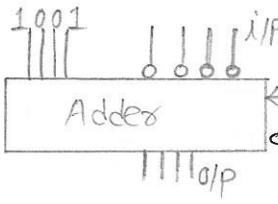
Binary to Gray.



gray to binary



BCD to exess-3



9's complement of BCD.

A/D C and D/A C

- Resolution = min. i/p voltage can be converted into digital.

$$= \frac{V_{ref}}{2^n}$$; n = bit of ADC/DAC

V_{ref} = Range of i/p supply.

- $V_{FS} = V_{ref} - [V_{(LSB)}]$

where; $V_{(LSB)}$ = step size = resolution

DAC o/p voltage general equ^n:-

$$V_0 = \frac{V_{ref}}{2^n}\left[b_3 2^3 + b_2 2^2 + b_1 2^1 + b_0 2^0\right]$$

for 4-bit DAC

ADC:-

1] Comparator type/Flash type/Parallel type.

- for n bit ADC. $(2^n - 1)$ o/p amp comparators are required.
- $(2^n \times n)$ priority encoder is also used.
- fastest ADC.

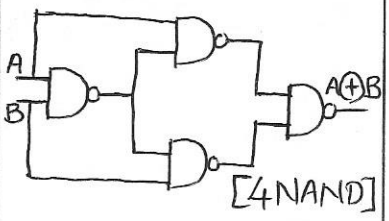
X-OR [A⊕B]

$Y = A'B + AB'$
$= (A+B)(A'+B')$

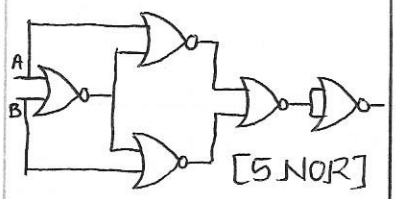
0 0	0
0 1	1
1 0	1
1 1	0

for odd no. of 1s in i/p

- $A \oplus B = [A \odot B]'$



[4 NAND]



[5 NOR]

$A \oplus B \oplus C = [(A \oplus B) \oplus C]$

$A \oplus 1 = A'$ $\quad A \odot 1 = A$

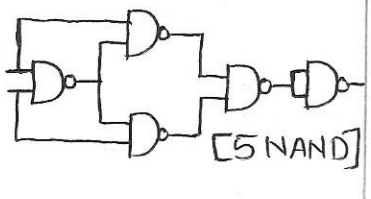
$A \oplus 0 = A$ $\quad A \odot 0 = A'$

X-NOR [A⊙B]

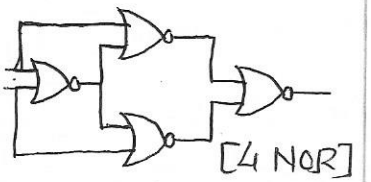
$Y = AB + A'B'$
$= (A'+B)(A+B')$

0 0	1
0 1	0
1 0	0
1 1	1

for even no. of 1s in i/p



[5 NAND]

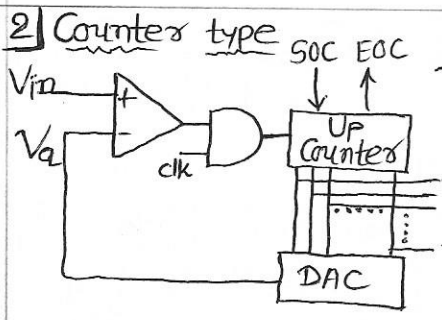


[4 NOR]

$A \odot B \odot C \neq [(A \odot B) \odot C]$

$[(A \odot B) \odot C] = A \oplus B \oplus C$ (for odd i/ps)

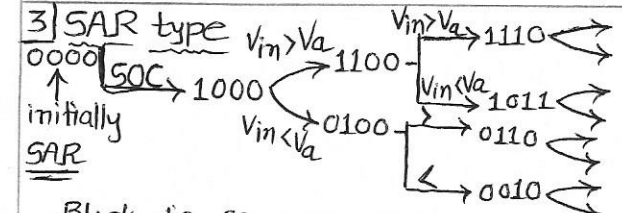
2] Counter type



- $V_{in} > V_a \Rightarrow$ op-amp = 1
  ↳ clk reaches to counter.
- counter increament by one.
- $V_{in} < V_a \Rightarrow$ opamp = 0
  ↳ Counter stops.

Max. Conversion time $= (2^n - 1) * t_{clk}$

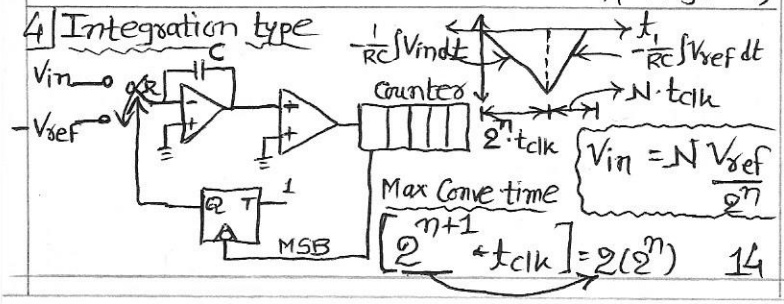
3] SAR type



- Block dia. same as counter type. iz replace counter by SAR resistor with control ckt.

Max. conversion time $= n * t_{clk}$ (independant of i/p magnitude)

4] Integration type



$V_{in} = \frac{N V_{ref}}{2^n}$

Max Conv time $[2^{n+1} * t_{clk}] = 2(2^n)$

K	M
A	P

→ gray code is used.

Min. Exp. :- $AB' + BC + A'B$ $\quad = AB' + AC + A'B$

Hazard free exp. :- $AB' + BC + A'B + AC$ $\quad = B'A + AC + A'B + BC$

Prime implicants [All the possible pairing options]

Essential Prime implicants (here = 2) [having one & only one possiblity of pairing]

- don't look for once pairing is done for higher no. of 1s.

(ex. :- don't consider 2 1s pairing if there is group of 4 1s).